

MEAN

The screenshot shows the Visual Studio Code interface with a project named 'DescriptiveStatistics'. The Explorer panel on the left shows a file named 'Mall_Customers.csv' and a Python file 'mean.py'. The main editor window displays the code in 'mean.py':

```
1 import pandas as pd
2
3 data = pd.read_csv("Mall_Customers.csv")
4
5 mean = data["Annual Income (k$)"].mean()
6
7 print("Mean =", mean)
```

The TERMINAL panel at the bottom shows the command prompt output:

```
PS C:\Users\hp\Downloads\DescriptiveStatistics> py mean.py
Mean = 60.56
PS C:\Users\hp\Downloads\DescriptiveStatistics>
```

The right sidebar contains a CHAT panel with a tip to try the 'Blan agent' and a 'mean.py' file icon.

SAMPLE MEDIAN

The screenshot shows the Visual Studio Code interface with the same project 'DescriptiveStatistics'. The Explorer panel now shows three files: 'Mall_Customers.csv', 'mean.py', and 'median.py'. The main editor window displays the code in 'median.py':

```
1 import pandas as pd
2
3 data = pd.read_csv("Mall_Customers.csv")
4
5 sample_median = data["Age"].median()
6
7 print("Sample Median =", sample_median)
```

The TERMINAL panel at the bottom shows the command prompt output:

```
PS C:\Users\hp\Downloads\DescriptiveStatistics> py median.py
Sample Median = 36.0
PS C:\Users\hp\Downloads\DescriptiveStatistics>
```

The right sidebar contains a CHAT panel with a tip to try the 'Blan agent' and a 'median.py' file icon.

SAMPLE VARIANCE

The screenshot shows a Visual Studio Code editor window with the file explorer on the left displaying a project named 'DescriptiveStatistics' containing 'Mall_Customers.csv', 'mean.py', and 'variance.py'. The 'variance.py' file is open in the editor, showing the following Python code:

```
1 import pandas as pd
2
3 data = pd.read_csv("Mall_Customers.csv")
4
5 sample_variance = data["Age"].var()
6
7 print("Sample Variance =", sample_variance)
```

The terminal at the bottom shows the command 'py variance.py' being executed, resulting in the output: 'Sample Variance = 195.13316582914572'.

STATUS BAR: Ln 7, Col 44 Spaces: 4 UTF-8 CRLF Python

COVARIANCE

The screenshot shows a Visual Studio Code editor window with the file explorer on the left displaying a project named 'DescriptiveStatistics' containing 'covariance.py', 'Mall_Customers.csv', 'mean.py', 'median.py', and 'variance.py'. The 'covariance.py' file is open in the editor, showing the following Python code:

```
1 import pandas as pd
2
3 data = pd.read_csv("Mall_Customers.csv")
4
5 covariance = data["Age"].cov(data["Annual Income (k$)"])
6
7 print("Covariance =", covariance)
```

The terminal at the bottom shows the command 'py covariance.py' being executed, resulting in the output: 'Covariance = -4.548743718592959'.

STATUS BAR: Spaces: 4 UTF-8 Python

CORRELATION

The screenshot shows a Visual Studio Code editor window with the file explorer on the left displaying a project named 'DescriptiveStatistics'. The file 'correlation.py' is open in the editor. The code in the file is as follows:

```
1 import pandas as pd
2
3 data = pd.read_csv("Mall_Customers.csv")
4
5 correlation = data["Age"].corr(data["Annual Income (k$)"])
6
7 print("Correlation =", correlation)
```

The terminal at the bottom shows the command 'py correlation.py' being executed, resulting in the output: 'Correlation = -0.012398042736060138'.

On the right side, there is a 'CHAT' panel with a prompt: 'Describe what to build'. Below the prompt are buttons for '+ Agent', 'Auto', and a search icon.

SUMMARIZATION

The screenshot shows a Visual Studio Code editor window with the file explorer on the left displaying a project named 'DescriptiveStatistics'. The file 'data_summarization.py' is open in the editor. The code in the file is as follows:

```
8 print(data.head())
9
10 print("\nDataset Shape:")
11 print(data.shape)
12
13 print("\nColumn Names:")
14 print(data.columns)
15
16 print("\nData Types:")
17 print(data.dtypes)
18
19 print("\nMissing Values:")
20 print(data.isnull().sum())
21
22 print("\nStatistical Summary:")
23 print(data.describe())
```

The terminal at the bottom shows the command 'py data_summarization.py' being executed, resulting in the output of a statistical summary:

```
Statistical Summary:
CustomerID      Age  Annual Income (k$)  Spending Score (1-100)
count  200.000000  200.000000          200.000000          200.000000
mean    100.500000   38.850000           60.500000           50.200000
std     57.879185   13.969007           26.264721           25.823522
min      1.000000   18.000000           15.000000            1.000000
25%     50.750000   28.750000           41.500000           34.750000
50%    100.500000   36.000000           61.500000           50.000000
75%    150.250000   49.000000           78.000000           73.000000
max    200.000000   70.000000          137.000000          99.000000
```

On the right side, there is a 'CHAT' panel with a prompt: 'Describe what to build'. Below the prompt are buttons for '+ Agent', 'Auto', and a search icon.